

COPY

ORIGINAL

ShinEtsu

MSDS NO: EU-06-05300717

1/7

PRODUCT NAME: KF-860

MATERIAL SAFETY DATA SHEET

SECTION 1. COMPANY IDENTIFICATION

COMPANY IDENTIFICATION

MANUFACTURER'S NAME: Shin-Etsu Chemical Co., Ltd.

ADDRESS: 6-1, 2-Chome, Ohtemachi, Chiyodaku, Tokyo, JAPAN

EMERGENCY TELEPHONE NUMBER:

743-7277 (Shin-Etsu Singapore Pte., Ltd.)

TELEPHONE NUMBER FOR INFORMATION:

03-3246-5121 (Tokyo, JAPAN)

743-7277 (Shin-Etsu Singapore Pte., Ltd.)

DATA PREPARED : 03/30/1994

LAST REVISION : 11/14/2008

DATA ISSUED : 10/21/2009

ISSUE NO 200910002077

BASE NO 10

PRODUCT NAME :

KF-860

PRODUCT CLASSIFICATION:

Silicone Fluid

SECTION 2. COMPOSITION

SINGLE OR MIXTURE:

Single

CHEMICAL IDENTIFICATION:

Modified organopolysiloxane

HAZARDOUS COMPONENT(S)/(CAS No.):

Modified organopolysiloxane/

(Confidential) [Combustible Liquid] : 100%

Octamethylcyclotetrasiloxane (impurity) /

(556-67-2) [Flammable Liquid] : ca. 2 %

(See Section 8 of this MSDS for Exposure Guideline)

(See Section 11 of this MSDS for Subacute toxicity and Reproductive effect.)

SECTION 3. HAZARDS IDENTIFICATION

HAZARDS CLASSIFICATION:

None (based on IMO)

Combustible Liquids (based on DOT)

FIRE AND EXPLOSION:

Combustible

POTENTIAL HEALTH EFFECT:



PRODUCT NAME: KF-860

INHALATION ; Mist irritating to the respiratory tract. Highly toxic by aerosol inhalation, may cause death.
SKIN contact ; May cause slight skin irritation.
EYES contact ; Causes eyes irritation.
INGESTION ; No information is available.

SECTION 4. FIRST AID MEASURES

INHALATION ; Remove to fresh air and promptly call a physician.
SKIN contact ; Remove product from skin with dry cloth or towel, and wash exposed area with detergent.
EYES contact ; Immediately flush with water for at least 15 minutes.
INGESTION ; Wash out mouth with water provided person is conscious. Never give anything by mouth to an unconscious person. Call a physician immediately.

SECTION 5. FIRE FIGHTING MEASURES

FLASH POINT (method used):

95 degrees C (Open cup)
>61 degrees C (Closed cup)

FLAMMABLE LIMITS:

LOWER: Not measured UPPER: Not measured

EXTINGUISHING MEDIA:

Foam, dry chemical or carbon dioxide

SPECIAL FIRE FIGHTING PROCEDURE:

None

UNUSUAL FIRE AND EXPLOSION HAZARD:

None

SECTION 6. ACCIDENTAL RELEASE MEASURES

STEP TO BE TAKEN IN CASE MATERIAL IS RELEASED OR SPILLED:

Shut off all ignition sources.
Contain the spill or leak.
Scrape up with cardboard or rag and place in container.

SECTION 7. HANDLING AND STORAGE

PRECAUTION TO BE TAKEN IN HANDLING AND STORING:

Keep container closed when not in use.
Store in a cool place.
Keep away from heat, sparks and flame.
Do not lay the container on its side.

Use with adequate ventilation.
Avoid contact with eyes and skin.
Do not breathe mist.
Keep out of reach of children.
* * * * * Information about the emptied container * * * * *
Do not re-use this container.
Residues will be explosive or flammable.
This container will be very hazardous when emptied.
Keep away from heat, sparks and flame.
Do not puncture or cut this container, and do not weld on or
near this container.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

EXPOSURE GUIDELINES:

ACGIH TLV-TWA : Not established,
OSHA PEL : Not established
[Octamethylcyclotetrasiloxane ; impurity]

RESPIRATORY PROTECTION(specify type):

Use respiratory protection unless adequate local exhaust
ventilation is provided.(Organic vapor type)

VENTILATION:

LOCAL EXHAUST: Recommended
MECHANICAL(general): Adequate ventilation system
SPECIAL: Unknown
OTHER: Unknown

PROTECTIVE GLOVES:

Plastic film or rubber gloves

EYE PROTECTION:

Safety glasses

OTHER PROTECTIVE CLOTHING OR EQUIPMENT:

Eyewash equipment

WORK/HYGIENIC PRACTICES:

Keep away from heat, sparks and flame.
Do not breathe mist.
Wash hands and gargle after handling.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

BOILING POINT:

Not applicable

VAPOR PRESSURE:

Negligible (25 degrees C)

VAPOR DENSITY(air=1):

Not applicable

SPECIFIC GRAVITY:

0.97 (25 degrees C)

MELTING POINT:

Not applicable

EVAPORATION RATE:

Negligible (Butyl Acetate=1)

SOLUBILITY IN WATER:

Not soluble

APPEARANCE(color):

Colorless, transparent

APPEARANCE(form):

Fluid

ODOR:Slight odor

SECTION 10. STABILITY AND REACTIVITY

STABILITY:

Stable

CONDITION TO AVOID:

None

INCOMPATIBILITY(material to avoid):

None

HAZARDOUS DECOMPOSITION OR BY-PRODUCT:

Thermal breakdown of this product during fire or very high heat condition may evolve the following hazardous decomposition product: Carbon oxides and traces of incompletely burned carbon compounds. Silicon dioxide. Nitrogen oxides. Formaldehyde.

HAZARDOUS POLYMERIZATION:

Will not occur

CONDITION TO AVOID:None

SECTION 11. TOXICOLOGICAL INFORMATION

SKIN IRRITATION:

Primary Skin Irritation-RABBIT : MILD

EYE IRRITATION:

No information is available.

ACUTE TOXICITY(LD50):

No information is available.

ACUTE TOXICITY(LC50):

In studies of rats exposed by inhalation to aerosols of specific aminofunctional silicones show highly toxicity (LC50:<0.5mg/1/4Hr). But for other routes(oral, ingestion and dermal), these responses have not been observed.

SUBACUTE TOXICITY:

Repeated inhalation or oral exposure of mice and rats to

octamethylcyclotetrasiloxane produced an increase in liver size. No gross histopathological or significant clinical chemistry effects were observed.

An increase in liver metabolizing enzymes, as well as a transient increase in the number of normal cells (hyperplasia) followed by an increase in cell size (hypertrophy) were determined to be the underlying causes of the liver enlargement. The biochemical mechanisms producing these effects are highly sensitive in rodents, while similar mechanisms in humans are insensitive.

CHRONIC TOXICITY:

A two year combined chronic and carcinogenicity assay was conducted on octamethylcyclotetrasiloxane. Rats were exposed by whole-body vapor inhalation 6hrs/day, 5days/week for up to 104weeks to 0, 10, 30, 150 or 700ppm of octamethylcyclotetrasiloxane. The increase in incidence of (uterine)endometrial cell hyperplasia and uterine adenomas (benign tumors) were observed in female rats at 700ppm. Since these effects only occurred at 700ppm, a level that greatly exceeds typical workplace or consumer exposure, it is unlikely that industrial, commercial or consumer uses of products containing octamethylcyclotetrasiloxane would result in a significant risk to humans.

CARCINOGENICITY:

NTP: Not listed, IARC: Not listed, OSHA REGULATED: Not listed

REPRODUCTIVE EFFECT:

Octamethylcyclotetrasiloxane administered to rats by whole body inhalation at concentrations of 500 and 700 ppm for 70 days prior to mating, through mating, gestation and lactation resulted in decreases in live litter size. Additionally, increases in the incidence of deliveries of offspring extending over an unusually long time period (dystocia) were observed at these concentrations. Statistically significant alterations in these parameters were not observed in the lower concentrations evaluated (300 and 70 ppm). In a previous range-finding study, rats exposed to vapor concentrations of 700 ppm had decreases in the number of implantation sites and live litter size. The significance of these findings to humans is not known.

TERATOGENIC EFFECT:

No teratogenic effects in the rats and rabbits
[Octamethylcyclotetrasiloxane].

OTHER INFORMATION:

None

SECTION 12. ECOLOGICAL INFORMATION

BIODEGRADATION:

No information is available.

BIOACCUMULATION:

No information is available.

AQUATIC TOXICITY:

No information is available.

OTHER INFORMATION:

None

SECTION 13. DISPOSAL CONSIDERATIONS

Can be burned in a chemical incinerator equipped with an
afterburner and scrubber.

Do not dispose of the emptied container unless the contents
have been completely removed and container has been flushed
with a clean neutral solvent and then dried up.

Do not dispose the emptied container unlawfully.

Observe all federal, state, and local laws.

SECTION 14. TRANSPORT INFORMATION

<IMO INFORMATION>**UN No.:**

None

CLASSIFICATION AND CLASS:

None

PACKAGING GROUP:

None

PROPER SHIPPING NAME:

None

TECHNICAL SHIPPING NAME:

None

MARINE POLLUTANT:

None

<DOT INFORMATION>**ID No.(49CFR 172.101):**

NA 1993

HAZARD CLASS(49CFR 172.101):

Combustible Liquid

PACKING GROUP(49CFR 172.101):

III

PROPER SHIPPING NAME(49CFR 172.101):

Combustible Liquids, N.O.S.

TECHNICAL SHIPPING NAME:

Organopolysiloxane

DOT REPORTABLE QUANTITY(49CFR 172.101, APP.) :**HAZARD SUBSTANCE(S) NAME / (CAS No.), CONTENTS AND RQ**

Not applicable

SECTION 15. REGULATORY INFORMATION

TOXIC SUBSTANCES CONTROL ACT(TSCA) STATUS:

Listed on the TSCA Inventory.

SUPERFUND AMENDMENTS AND REAUTHORIZATION ACT OF 1986(SARA)**TITLE III SECTION 313 SUPPLIER NOTIFICATION:**

This regulation requires submission of annual reports of toxic chemical(s) that appear in section 313 of the emergency planning and community Right-To-Know Act of 1986 and 40 CFR 372.

This information must be included in all MSDS's that are copied and distributed for the material.

The toxic chemical(s) contained in this product are:

CHEMICAL NAME/(CAS No.) AND CONTENTS

** None **

CALIFORNIA PROPOSITION 65:

This regulation requires a warning for California Proposition 65 Chemical(s) under the statute.

The California Proposition 65 Chemical(s) contained in this product are:

CHEMICAL NAME/(CAS No.) AND CONTENTS

** None **

SECTION 16. OTHER INFORMATION

For Industrial Use Only

Do not use this product for consumers' spray. Highly toxic by aerosol inhalation, may cause death. Use only with adequate ventilation, wear appropriate protective equipment, including approved respirator.

This materials safety data sheet is offered solely for your information, consideration and investigation.

The data described in this MSDS consist of data on literature, our acquisitional data and analogical inference by data of similar chemical substance or product.

Shin-Etsu Chemical Co., Ltd. provides no warranties, either express or implied, and assumes no responsibility for the accuracy or completeness of the data contained herein.
